# GATESYNC MANAGEMENT PLATFORM

## SMART RESIDENTIAL SOCIETY & GATE SECURITY DESK — PROJECT REPORT

P P Savani University | B.Tech Information Technology & Engineering (7th Sem)

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## 1. Executive Abstract & System Overview

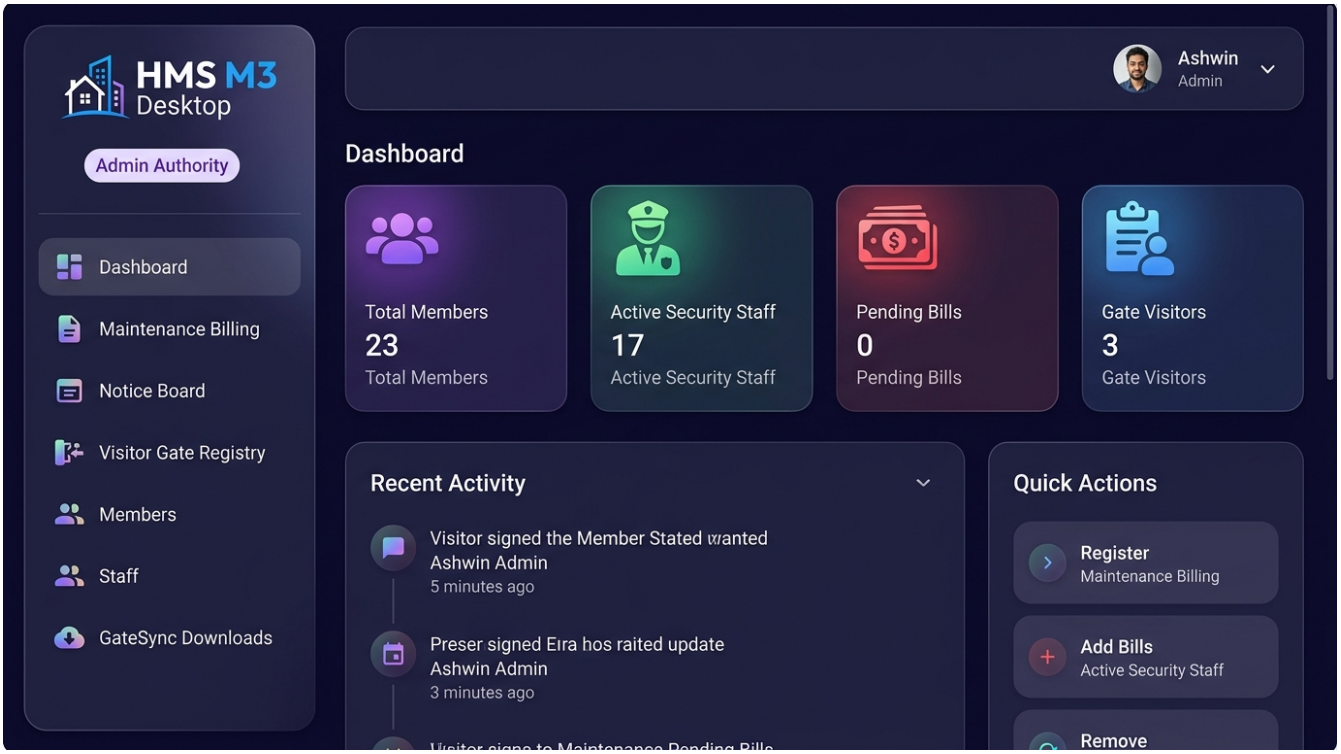
GateSync (Gate & Society Management System) is a high-performance web platform engineered using React 19 and Google Material Design 3. The platform automates housing society governance, maintenance billing calculations, gate security visitor logging, notice broadcasts, and financial data exports.

## 2. Core System Objectives

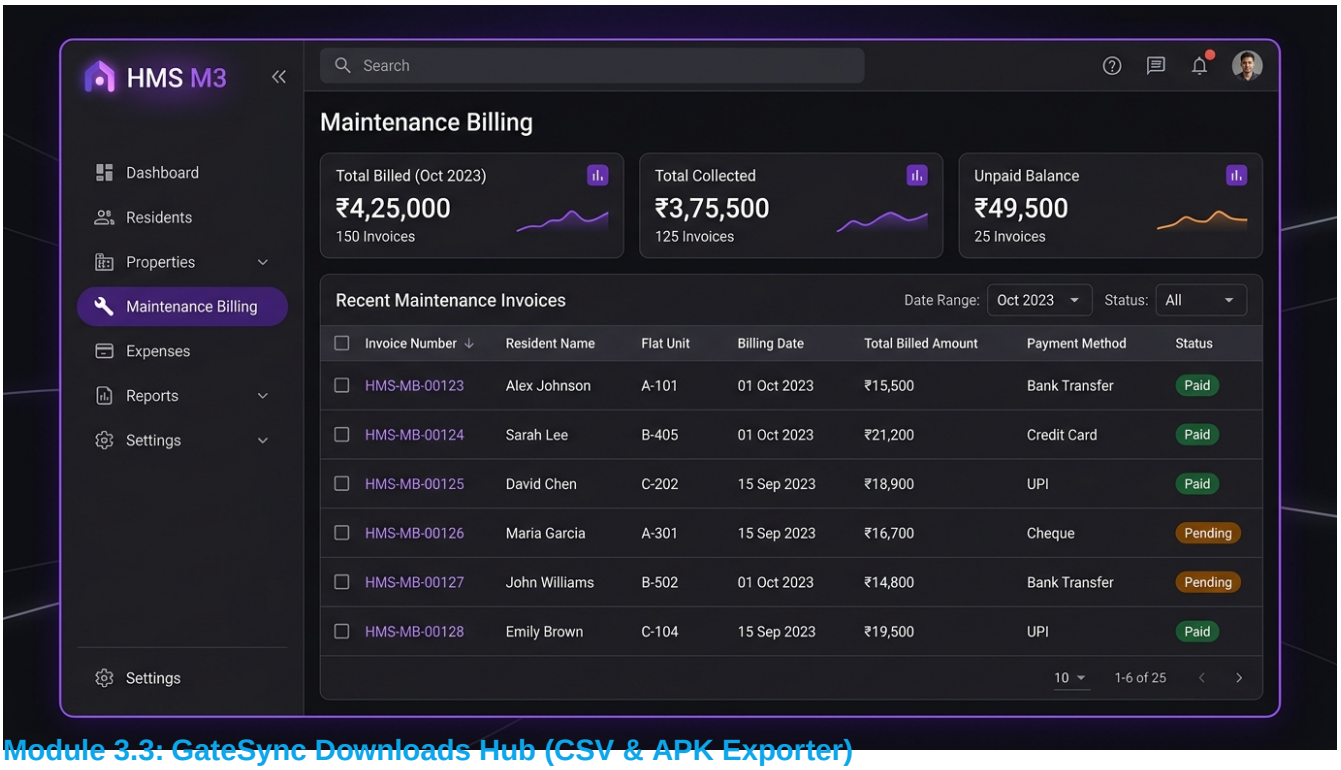
- Automate Maintenance Billing: Dynamic itemized bill generation (Block Maintenance, Electricity, Water).
- Role-Based Portal Access: Distinct UI control desks for Admin Authority, Resident Portal, and Security Gate.
- Security & Gate Registry: Real-time visitor entry/exit tracking with phone validation and privacy filters.
- GateSync Downloads Hub: Direct CSV exports for visitor logs, maintenance ledgers, and resident directories.

## 3. UI Interface & Module Specifications

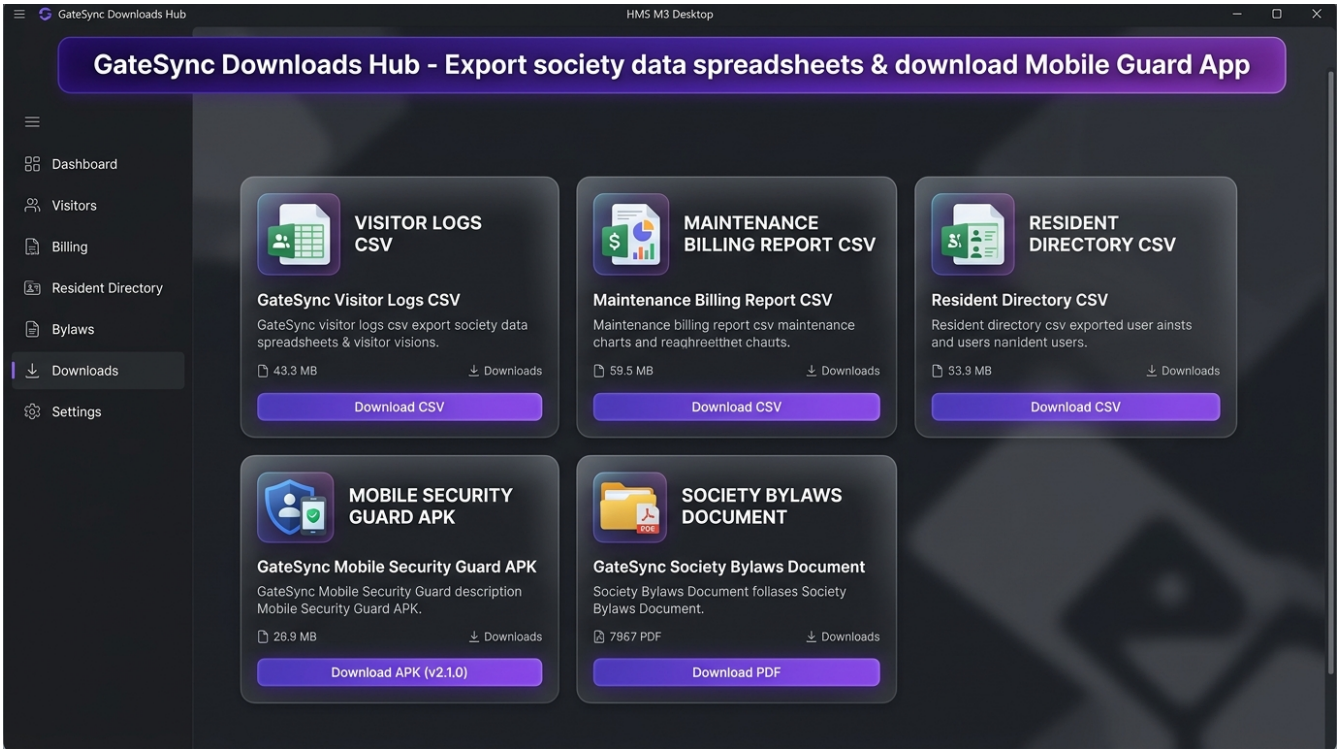
### Module 3.1: Executive Dashboard Desk



## Module 3.2: Maintenance Billing & Payment Ledger




## Module 3.3: GateSync Downloads Hub (CSV & APK Exporter)



## 4. Codebase Architecture & Build Verification

The application codebase is structured under src/ into components, context, data models, and views. State management is powered by AppContext with localStorage state persistence.

### Build Quality & Testing Verification Matrix

| Test Case | Command | Result | Status |
|---|---|---|---|
| Vite Build | npm run build | Transformed 1606 modules | PASSED |
| Dev Server | npx vite --port 5173 | Running on localhost:5173 | PASSED |
| Data Exporter | CSV Export Handler | Generates formatted CSVs | PASSED |

## 5. Conclusion & Future Scope

GateSync delivers an end-to-end digital governance solution. Future roadmap plans include real-time WebSocket gate push alerts, online payment gateway reconciliation, and automated camera license plate reading.